Exposing and Addressing Fake Base Station Vulnerabilities in 5G Through User Device Exploits

Yi-Hsueh Tsai and Shiang-Jiun Chen, *National Taipei University of Technology, Taipei, Taiwan*

The rapid advancement of 5G networks introduces new security challenges, particularly with the rise of false base station (FBS) attacks. This study investigates the vulnerabilities of 5G networks exploited by FBSs, which hijack communications by mimicking legitimate base stations and compromising user equipment (UE). Through simulated experiments, we demonstrate how attackers can capture and replay 5G authentication vectors, bypassing security protocols and gaining unauthorized access. Key attack vectors include man-in-the-middle attacks and false mobility management function replays. Our findings highlight the need for enhanced authentication protocols and real-time detection systems to mitigate these threats. This research provides critical insights into securing 5G networks, emphasizing the importance of adaptive defense strategies against evolving cyber threats.

The introduction of 5G technology has marked a significant transformation in telecommunications, offering capabilities such as ultrareliable low-latency communications, enhanced mobile broadband, and massive machine-type communications. These advancements, however, bring a host of new security challenges, especially due to the exponential increase in connected devices. As outlined in recent research, 5G networks are susceptible to various threats, including compromised user equipment (UE), man-in-the-middle (MitM) attacks, and the use of fake base stations (FBSs).[1,2]

The 5G network architecture, which heavily relies on software-defined networking and network function virtualization, adds layers of complexity. Vulnerabilities in one element can easily propagate throughout the system, exacerbating the risk of security breaches. FBS attacks, the focus of this study, take advantage of these architectural complexities by mimicking legitimate base stations to hijack communications and bypass authentication protocols.[3] This makes 5G networks particularly vulnerable to such attacks, posing significant risks to user privacy and network security.

Mobile network advancements have consistently brought increased speed and new security risks over the generations, from 2G to 5G. The unique structure of 5G networks, which involves complex software and hardware integrations, heightens the difficulty of isolating and segmenting network functions. This complexity allows attackers to exploit cross-network vulnerabilities, even in private 5G deployments, where the expectation of isolated environments is often compromised.[4,5]

Given the critical role 5G networks will play in future industries, including private and industrial environments, addressing these security risks is crucial. Enterprises and organizations deploying 5G must adopt enhanced security measures, perform regular inspections, and update systems to mitigate these emerging threats.[6]

While this study addresses vulnerabilities within the current 5G framework, it also acknowledges the potential of the Zero Trust model. Zero Trust, based on

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the principle of "never trust, always verify," enforces strict identity verification, continuous monitoring, and granular access controls. Implementing Zero Trust would require a fundamental redesign of 5G networks, including changes to authentication processes and network segmentation. This research focuses on enhancing security within the existing 5G architecture while recognizing Zero Trust as a promising future direction.

The multilayer architecture, network software virtualization, multiconnectivity, and high-speed data transmission characteristics of 5G significantly expand the attack surface. These features enable adversaries to conceal malware within normal traffic or intercept data streams, exploiting the inherent complexities of the system. False Base Station (FBS) attacks leverage these attributes to amplify vulnerabilities and execute rapid attacks, as mentioned in 3GPP TR 33.809-i01.[7]

This article explores the vulnerabilities associated with FBS attacks in 5G networks, focusing on how attackers exploit authentication weaknesses. This study demonstrates how malicious actors can capture and replay authentication vectors (AVs) through simulated FBS attacks, gaining unauthorized access to 5G networks. Our findings highlight the importance of improving authentication protocols and deploying robust detection systems to prevent such attacks. The research contributes to ongoing efforts to strengthen 5G network security and offers insights into advanced mitigation strategies to protect against evolving cyberthreats.[8,9,10]

RELATED WORK

The 5G security framework addresses various threats, including radio access signaling and user plane data breaches, confidentiality leaks, impersonation, replay attacks, MitM intrusions, and downgrade attacks. These vulnerabilities were carefully examined during the design and deployment phases of 5G networks to ensure strong defenses against potential security risks. This approach highlights the importance of security in maintaining the integrity and reliability of 5G communications, ultimately safeguarding both users and operators.[11]

5G Cybersecurity

Mohan et al.[5] note that the rise in network connectivity, increased user numbers, and the absence of standardized Internet of Things protocols significantly heighten security threats. In the context of 5G, network attacks are categorized into three types: physical, remote, and local. Each category comprises a range of vectors, including hardware, operational, and network-related vulnerabilities. Addressing these attacks requires handling diverse threats, such as denial-of-service, distributed denial-of-service, spoofing, MitM, jamming, eavesdropping, data sniffing, and Sybil attacks.

Jover et al.[12] highlight several flawed assumptions and protocol loopholes in the 5G system that could make communication networks vulnerable to attacks. For example, the continued support for null encryption and authentication can be exploited. The 5G standard must ensure that no insecure protocol edge cases, such as null authentication, null encryption, downgrade attacks, and exploitation of preauthentication messages, are permitted in any operational mode. Additionally, 3GPP TR 33.809-i01[7] addresses user privacy and network availability challenges.

According to NIST's Special Publication 1800-33,[13] previous LTE systems required UE to respond with cleartext International Mobile Subscriber Identity, exposing the UE's identity to rogue base stations. In 5G, using a subscription permanent identifier (SUPI) and the subscriber concealed identifier enhances privacy, provided that the correct security features are enabled. However, users accessing 5G services through non-3GPP networks, such as Wi-Fi, remain vulnerable to attacks like session hijacking. Therefore, a secure connection context must be maintained across trusted and untrusted networks, with the Non-3GPP Interworking Function (N3IWF) facilitating secure access via IPsec tunnels.

Salim et al.[14,15] highlight a vulnerability in the General Packet Radio Service (GPRS) Tunnel Protocol (GTP) used for transmitting data between the 5G core (5GC) and base stations. The protocol lacks encryption, exposing the 5GC to attacks. Additionally, vendors often do not implement packet verification mechanisms, creating potential security risks.

Dolente et al.[16] assess vulnerabilities in two externally exposed core network functions—the access and mobility management function (AMF) and the network repository function/network exposure function—under the service-based architecture. Their findings emphasize the importance of rigorous security assessments to ensure the safety of these critical components.

FBSs

FBSs, known as "base station spoofing," involve attackers setting up counterfeit base stations to intercept or hijack mobile device connections. This attack

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compromises user privacy, allowing attackers to steal personal information or monitor communications. By mimicking legitimate base stations, attackers deceive devices into connecting, posing a significant threat to 5G networks.

Sun et al.[17] propose an artificial intelligence/machine language-based detection method that integrates temporal graph analysis with anomaly detection algorithms to enhance FBS detection accuracy. By leveraging UE connection events and wireless measurement reports, they deploy isolation forest and local outlier factor algorithms to identify anomalies, minimizing false positives.

Karaçay et al.[18] introduce a network-based localization method to estimate the precise position of FBSs using reference signal received power values. Their experimental results using the ns-3 simulator demonstrate that this method can locate FBSs with meter-level accuracy in various scenarios.

Zhang et al.[19] analyze the FBS spam ecosystem, identifying business categories and patterns in spam messages. Their research highlights the organizational structure of FBS attacks and how they exploit marketing strategies to deceive users while evading detection.

FBS attacks often disrupt UE services, enabling attackers to manipulate communication between the UE and the network. Attackers may relay or modify unprotected messages while transmitting secure ones without alteration. The level of mimicry depends on the attacker's objectives. In some cases, attackers may not closely imitate legitimate base stations but can still launch effective attacks by exploiting weaknesses in detection mechanisms.

Traditional Defense Mechanisms

Traditional defenses against FBS attacks typically involve modifying wireless communication protocols, strengthening authentication processes, or adding encryption layers. In previous generations of wireless technology, systems were designed to use public key infrastructure and encryption protocols to secure communications

EXPERIMENTATION FOR FALSE BASE STATION

This section will introduce the test framework, environment, and scenario for FBSs; the Amarisoft UE, Nemo Handy UE, and Amarisoft gNodeB (gNB) are primarily used to reproduce the attack scenarios within a simulated environment, which does not entirely reflect the real-world operation of 5G radio access network (RAN). In a real 5G operational environment, the UE is expected to verify the correctness of the sequence number (SQN) in the AV and reject the authentication response if anomalies are detected. However, the test framework in this setup might not incorporate such rigorous verification mechanisms in the simulated UE to demonstrate how attackers can exploit these vulnerabilities to hijack communications successfully.[20]

Test Framework for FBSs

The test framework developed for this experiment aims to investigate the vulnerabilities of 5G networks when targeted by an FBS attack. The experiment simulates a real-world scenario in which a malicious actor deploys a counterfeit base station and core network to hijack UE. The framework utilizes Amarisoft UE, Nemo Handy UE, and an Amarisoft gNB. This gNB handles the RAN functions, providing the necessary interface for communication between the UE and the core network.

The core network environment consists of multiple elements, such as the access and mobility management function (AMF), the unified data management (UDM), and the session management function, all working together to support the network's operations. These components are integrated with a Python-based 5G Core (5GC) emulator to simulate the operator's infrastructure. The emulators are developed using P1 Security's pycrate library, a Python library designed for manipulating various digital formats, particularly those related to cellular network signaling.

This framework aims to demonstrate how an attacker can capture and replay the 5G authentication vector (AV) to trick the victim UE into connecting to the FBS. The FBS setup leverages these core elements to establish communication with the victim UE. In addition to the legitimate core network, a separate Python-based 5GC emulator is implemented to simulate the operations of a rogue core network. This false network includes its AMF and UDM, which handle the replayed AVs and attempt to hijack the UE.

In Figure 1, the interface N2 handles the control-plane communication between the base station and the core network, specifically between the gNB and the AMF. The fake network extracts critical information such as the 5G AV, which includes parameters like random number (RAND), authentication token (AUTN), and SUPI. These extracted values are crucial for replaying the authentication process in the FBS setup.[20]

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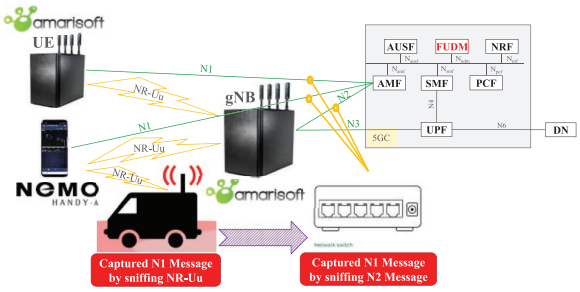

FIGURE 1. Framework for intercepting 5G control-plane communication: extraction of AVs.

Experimental Identification of 5G Network Vulnerabilities: Addressing the Ambiguity of Out-of-Range SQN in Security Testing

In the FBS test framework, after successfully capturing the AV and receiving the registration request from the UE (step ① of Figure 2), the false AMF sends an authentication request (step ② of Figure 2) with the replayed AV to the victim UE. If the UE does not detect any abnormalities—such as an SQN in the AV being out of range—it responds with an authentication response (step ③ of Figure 2), allowing the attacker to bypass the authentication phase. The false AMF sends a security mode command (SMC) (step ④ of Figure 2) with null security settings. If the UE accepts the null configuration, it sends a security mode complete message (step ⑤ of Figure 2), and the false base station gains full control over the UE's network entry after the AMF receives the registration complete message (step ⑥ of Figure 2) from the UE. This framework demonstrates how a sophisticated attacker could exploit these vulnerabilities to compromise 5G network security.[20]

If all of the access stratum (AS) and nonaccess stratum (NAS) messages are captured during the registration process, from the start to completion, the FBS could replay these AS and NAS messages to the victim's UE. Since the attacker can replay the entire registration sequence, there would be no need to send an SMC with null security settings to carry out the attack.

As described in the previous paragraph, the FBS attack can exploit the captured 5G AV to hijack the UE's communication. Using Amarisoft UE, Nemo Handy UE, and Amarisoft gNB in the test framework, the attacker can simulate this scenario, showing how the attack could be executed without relying on sending an SMC with null security settings. Instead, the replay of the captured messages itself is sufficient.

In a real-world 5G operational environment, the Universal Subscriber Identity Module (USIM) in the UE is expected to verify that the SQN is within the correct range, as defined in section 7.3.1.2 of 3GPP TS 31.122[21] to prevent replay attacks. However, in a simulated test environment, where such rigorous verification may not be correctly implemented due to the ambiguity of out-of-range SQNs defined in section 7.3.1.2 of 3GPP TS 31.122 for security testing, an attacker can successfully replay the AV to bypass authentication. If the UE fails to detect abnormalities in the AV, it will respond with an authentication response, allowing the attacker to bypass the normal authentication phase.

This scenario demonstrates that when the attacker can fully capture and replay the message flow during the registration process, they do not need to use an SMC with null security settings. The FBS attack can still hijack the UE by replaying the AS and NAS messages, effectively circumventing the 5G security protocols and compromising the network.

Framework for Assessing 5G Vulnerabilities Through Experimentation: Clarifying the Correct Range for SQN in Security Testing

In this scenario, first, the registration request (step ① of Figure 3) is intercepted to extract the value of SUPI. Next, the authentication request (step ② of Figure 3) is sniffed to extract the values of RAND and AUTN in the 5G AV. Finally, the authentication response is captured to extract the value of RES (step ③ of Figure 3).[20]

The experimentation environment also incorporates network sniffing tools to capture the necessary N1 and NAS messages for the attack. This setup allows us to intercept and analyze the communication between the gNB and the UE, enabling the extraction of critical authentication information such as the 5G AV.

The captured data are stored in a database on the false UDM, from which it can be replayed during the FBS attack. This environment demonstrates how easily an attacker can manipulate 5G network operations and compromise the integrity of a 5G network[20] by sending an authentication request (step ⑤ of Figure 3) with the replayed AV to the victim UE.

The environment is further validated by conducting multiple penetration tests to assess the network's resilience to FBS attacks. These tests highlight weak points in the 5G authentication process and

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demonstrate how attackers can exploit vulnerabilities in real-world scenarios (steps ① to ③ of Figure 3). They also illustrate how to mitigate replay attacks (steps ④ to ⑥ of Figure 3) if the USIM in the UE correctly implements the mechanism to verify that the received SQN is within the correct range, as defined in section 7.3.1.2 of 3GPP TS 31.122.[21] Overall, the environment setup provides a controlled yet realistic platform for assessing the security risks posed by FBS attacks in 5G networks.

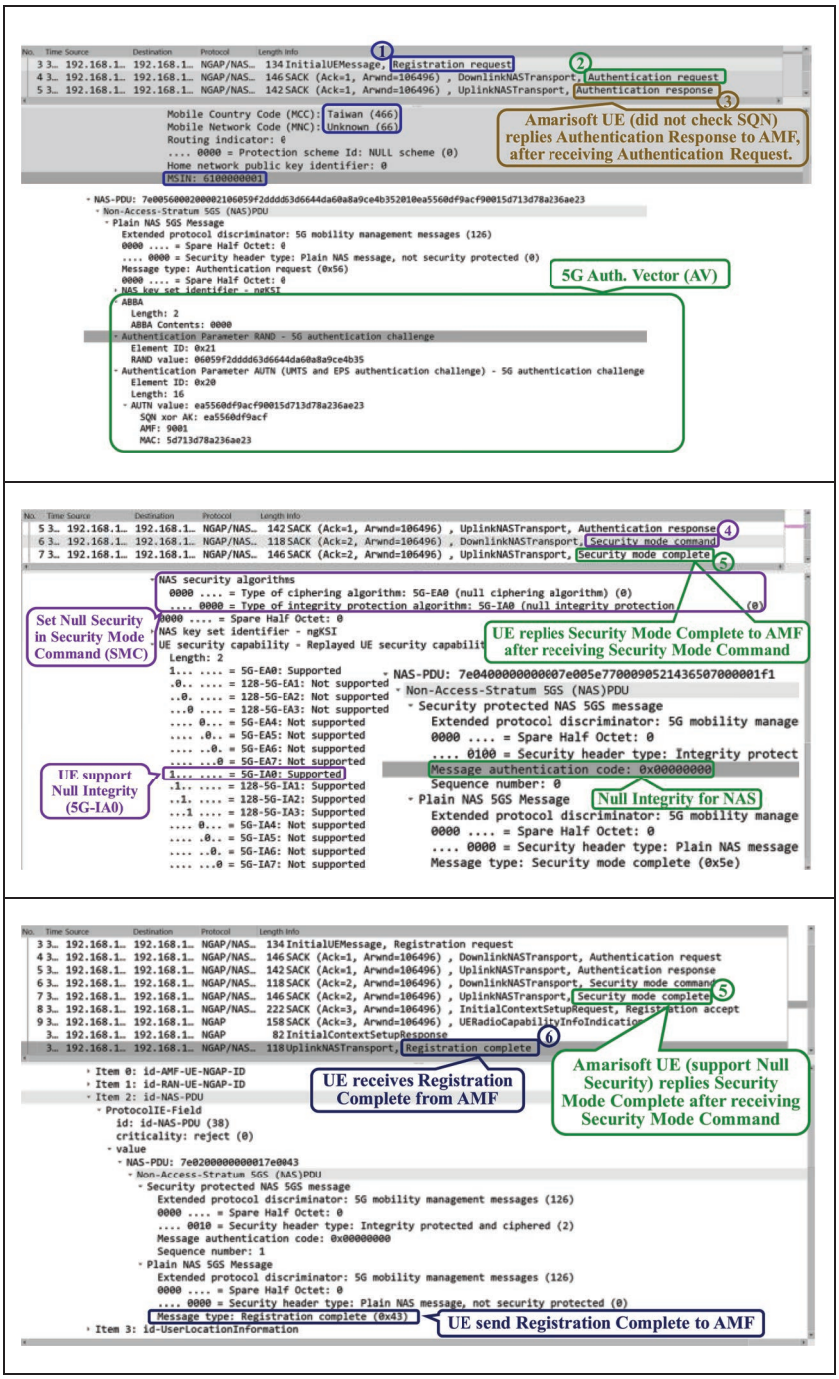

FIGURE 2. Replay Attack mechanism in 5G authentication.

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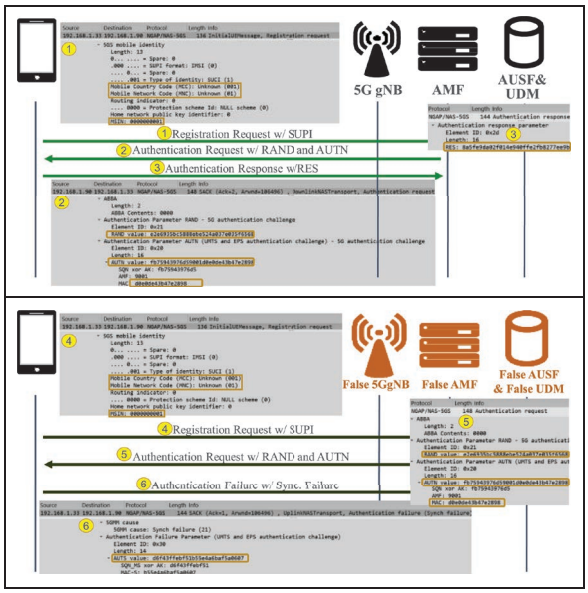

FIGURE 3. Extraction of 5G AV components and replay attack.

CONCLUSION

This study has revealed the substantial threat of FBSs to 5G networks, particularly in private and industrial environments. FBS attacks compromise communication security, exploit user privacy, and exploit weaknesses in network authentication protocols. Through detailed simulation experiments, we demonstrated how attackers can hijack UE by exploiting authentication vulnerabilities and bypassing core security mechanisms in 5G networks.

To mitigate these vulnerabilities, it is essential to implement more robust security measures, such as improved detection systems, enhanced authentication protocols, and continuous monitoring strategies. Leveraging machine learning techniques, particularly for anomaly detection, can significantly improve the detection of FBS activity and minimize the risk of widespread breaches. These measures align with global standards and recommendations from the Global System for Mobile Communications Association, emphasizing adaptive defense and proactive threat mitigation.

In conclusion, FBS attacks pose an evolving and critical security risk for 5G networks. Future efforts should focus on refining detection methodologies, enhancing security protocols, and adopting industry-wide standards to bolster network resilience. Network operators must prioritize real-time threat responses and implement advanced monitoring systems to protect against these sophisticated attacks. As the deployment of private 5G networks expands, ensuring robust cybersecurity measures is vital for safeguarding these crucial infrastructures.

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YI-HSUEH TSAI is an assistant professor at the National Taipei University of Technology, Taipei, 106344, Taiwan. His research interests include 5G and 6G mobile networks, nonterrestrial networks, and global positioning systems. Tsai received his Ph.D. degree from National Taiwan University. Contact him at yihsuehtsai@g.ntu.edu.tw.

SHIANG-JIUN CHEN is an assistant professor in the Department of Computer Science and Information Engineering at the National Taipei University of Technology, Taipei, 106344, Taiwan. Her research interests include numerical analysis, software-defined networking, and software testing. Chen received her Ph.D. degree from the University of Texas at Arlington. She is a Member of IEEE. Contact her at annette@ntut.edu.tw.

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